SELLVION

Combined Product Requirements Pack

BRD • DRD • SRS

Business Requirements • Design Requirements • Software Requirements

Draft for review | 12 November 2026

Prepared from the three SELLVION concept screenshots provided in this conversation. Items marked TBD or requiring validation remain open decisions and should be confirmed by the relevant stakeholders before implementation.

BRD — Business Requirements Document

SELLVION

Business Requirements Document

BRD | 0.1 — Draft for review | 12 November 2026

Source basis: three user-provided SELLVION concept screenshots.

Document status	Draft — requires stakeholder validation
Product	SELLVION — AI customer intelligence and engagement platform
Audience	Business stakeholders, product/design, engineering, QA and AI implementation teams

1. Executive summary

SELLVION is proposed as an autonomous AI employee for small and growing Indian e-commerce merchants. It uses available customer, order, purchase-history, preference, business-record and payment/udhaar information to identify opportunities and recommend or execute controlled customer-engagement actions. The business intent is to recover missed sales, improve repeat purchases and retention, reduce avoidable COD returns, simplify collections and reduce repetitive manual follow-up.

2. Business context and problem statement

The concept screenshots describe merchants losing customer continuity because records are scattered across notebooks and systems, inactive customers are hard to identify, reorder signals are missed, payment reminders are manual, and results are not measured. The screenshots cite an indicative India e-commerce COD return-to-origin range of 20–30% versus 2–3% for prepaid orders. These figures are supplied by the concept material and must be independently validated before being used in external claims or financial forecasts; they do not represent all merchant transactions.

3. Business objectives and expected outcomes

Objective	Expected business outcome	Measurement direction
Revenue growth	Recover missed sales and increase repeat purchases	Incremental repeat orders and revenue versus a control group
Customer retention	Bring inactive customers back	Reactivation rate and repeat-purchase rate
COD risk reduction	Identify risky orders and reduce avoidable returns	RTO/failed-delivery rate by eligible cohort
Faster collections	Simplify udhaar tracking and payment reminders	Overdue balance, collection rate and days-to-collect
Higher productivity	Reduce manual work and repetitive follow-ups	Staff time per customer/order and tasks automated
Personalised experience	Send relevant offers and timely reminders	Engagement, conversion and opt-out rates
Smarter decisions	Turn customer data into actionable insights	Action adoption and measurable outcomes
Affordable AI	Make intelligent automation accessible to small merchants	Cost-to-serve and merchant adoption

4. Target users and stakeholders

• Primary customer: small and mid-sized Indian e-commerce / retail merchant.

• Merchant owner or administrator: configures the business, integrations, policies, approval limits and reporting.

• Sales / customer-service / collections staff: review customer context, approve actions and manage exceptions.

• End customer: receives relevant WhatsApp or other approved messages, reorder reminders and payment follow-ups.

• SELLVION operations and support: manage onboarding, reliability, security, audit and customer support.

5. Proposed scope

5.1 In scope

• Customer intelligence: build customer profiles from permitted purchase histories, orders, preferences and business records.

• Behaviour analysis: identify inactive customers, regular buyers, changing purchase patterns and possible reorder opportunities.

• Smart predictions: surface reorder opportunities, outstanding udhaar and potential COD risk using explainable indicators.

• Multi-agent orchestration: coordinate specialised analysis, decision and execution agents under a central orchestrator.

• Automated engagement: draft or send personalised Hinglish WhatsApp messages, reorder reminders and payment follow-ups through approved providers.

• Earned autonomy: begin with merchant approval and gradually automate only repeatedly approved actions within merchant-defined limits.

• Revenue measurement: track responses, repeat purchases and estimated incremental revenue using a control-group approach.

• Digital ledger and billing capability is represented as a comparative capability in the screenshot; exact scope and integration boundary require confirmation.

5.2 Out of scope pending confirmation

• Replacing a full e-commerce storefront, payment gateway, shipping carrier or accounting platform.

• Guaranteeing delivery outcomes, COD success, revenue lift or a specific predictive accuracy.

• Sending messages without valid consent, compliant templates or provider approval.

• Autonomous refunds, order cancellations, credit decisions or financial transfers.

• Any specific ERP, CRM, commerce, billing or logistics integration not yet selected.

6. Differentiation and positioning

The concept positions SELLVION as one AI teammate connecting customer intelligence, predictive decisions, autonomous actions and measurable revenue outcomes. The comparison screenshot claims SELLVION coverage across digital ledger/billing, WhatsApp automation, customer behaviour analysis, predictive reorder intelligence, COD risk prediction, multi-agent orchestration, customer relationship memory graph, earned autonomy with owner control and incremental revenue measurement. Competitor capability marks are screenshot-provided positioning claims, not independently verified product research; validate them before publication.

7. Business requirements

ID	Business requirement	Priority
BR-01	The platform shall consolidate permitted customer and order information into a merchant-scoped customer view.	Must
BR-02	The platform shall identify customer segments including inactive, regular and changing purchasing behaviour.	Must
BR-03	The platform shall surface reorder opportunities and explain the signals behind each suggestion.	Must
BR-04	The platform shall track outstanding udhaar/payment follow-ups where source data is available.	Must
BR-05	The platform shall estimate COD risk and make the basis and uncertainty visible.	Should
BR-06	The platform shall create personalised engagement drafts and support approved delivery channels, including WhatsApp subject to provider rules.	Must
BR-07	The platform shall support merchant approval, configurable limits and an audit trail for automated actions.	Must
BR-08	The platform shall measure engagement, repeat purchases and incremental outcomes against a control group where feasible.	Must
BR-09	The platform shall provide dashboards for business outcomes and operational exceptions.	Should
BR-10	The platform shall provide a clear onboarding path suitable for small merchants.	Should

8. Success measures / KPIs

• Incremental revenue and repeat-order lift versus a defined holdout/control group.

• Inactive-customer reactivation rate; repeat-purchase frequency; reorder conversion.

• COD RTO rate for eligible orders, segmented by risk band and compared with baseline.

• Collection rate, overdue balance and average time to collect udhaar.

• Message delivery, response, conversion, opt-out and complaint rates.

• Percentage of recommendations approved, rejected, edited and successfully completed.

• Manual minutes saved per merchant per week; active merchant retention; platform cost per merchant.

9. Assumptions, constraints and dependencies

• Merchants can lawfully provide access to customer/order/business records and have the necessary rights and permissions.

• Data quality, identity matching and availability vary by merchant; incomplete records must be surfaced rather than silently inferred.

• WhatsApp delivery depends on an approved Business Platform/provider, templates, consent and applicable policy.

• Predictive models require sufficient historical data, ongoing monitoring and merchant-specific calibration.

• Merchant-defined approval thresholds, message frequency caps and quiet hours must be configured before automation.

• Indian privacy, consumer-protection, messaging, payment and e-commerce obligations require legal review.

10. Risks and mitigations

Risk	Mitigation
Poor or duplicated source data	Data validation, confidence labels, deduplication and exception queue
Incorrect COD/reorder prediction	Human review, model monitoring, explanations and conservative rollout
Unwanted messaging / consent failure	Consent records, opt-out handling, rate limits and provider-compliant templates
Automation causes customer or financial harm	Approval-first defaults, action allowlists, limits, reversibility and audit logs
Unclear revenue attribution	Holdout cohorts, baseline definitions and transparent estimation methodology
Integration outages	Retries, idempotency, queueing, alerts and manual fallback

11. Open decisions requiring stakeholder sign-off

• Confirm initial merchant segment, verticals and launch geography.

• Confirm whether DRD means Design Requirements Document (used in this pack) or Data Requirements Document.

• Confirm first-release scope for digital ledger/billing and the source of truth for udhaar balances.

• Select commerce, billing, logistics, payment and WhatsApp integration providers.

• Define supported customer channels, language policy and message approval process.

• Define user roles, approval thresholds, autonomy stages and restricted actions.

• Approve KPI formulas, control-group design and reporting windows.

• Confirm deployment model, data retention, residency, security and compliance requirements.

12. Acceptance of this BRD

This draft becomes a baseline only after business owner, product owner, engineering lead, operations and legal/privacy stakeholders review the open decisions and approve scope, priorities and measurable acceptance criteria.

DRD — Design Requirements Document

SELLVION

Design Requirements Document

DRD | 0.1 — Draft for review | 12 November 2026

Source basis: three user-provided SELLVION concept screenshots.

Document status	Draft — requires stakeholder validation
Product	SELLVION — AI customer intelligence and engagement platform
Audience	Business stakeholders, product/design, engineering, QA and AI implementation teams

1. Purpose and design principles

This document translates the SELLVION concept into a solution-design baseline for product, UX, data, AI, integration and engineering teams. It assumes DRD means Design Requirements Document. The architecture below is a proposed design, not a confirmed implementation.

• Merchant-first: every screen, record, model and action is scoped to a merchant tenant.

• Human control by default: recommendations precede automation; autonomy is earned through repeated approval and measurable reliability.

• Explainable decisions: show the evidence, confidence, freshness and reason for each insight.

• Safe-by-design engagement: consent, opt-out, quiet hours, frequency caps and provider rules are enforced before send.

• Composable agents: specialist agents have narrow permissions; an orchestrator coordinates but does not bypass policy controls.

• Outcome-oriented: capture action and outcome events so revenue and productivity impact can be measured.

2. Logical solution architecture

Layer	Proposed components	Design requirement
Experience	Merchant web console; mobile-responsive views; customer timeline; approval inbox; KPI dashboard	Role-aware, accessible, clear status and explanations
Application/API	Authentication, tenant management, customer/order APIs, workflow and notification APIs	Versioned APIs, validation, tenant isolation and idempotency
Data ingestion	Connectors, imports, webhooks, scheduled sync and reconciliation	Track source, timestamp, consent/legal basis, provenance and sync status
Data platform	Operational database, event store, analytics store, customer identity graph	Canonical IDs, deduplication, retention controls and lineage
Intelligence	Segmentation, reorder scoring, COD risk scoring, collections prioritisation	Versioned models, confidence, reason codes and monitoring
Agent orchestration	Customer analyst, prediction, message drafting, policy and execution agents	Least privilege, tool allowlists, approval gates and bounded execution
Engagement	WhatsApp Business provider adapter; future channel adapters	Template validation, consent checks, opt-out and delivery receipts
Governance	Audit log, policy engine, observability, security and admin controls	Tamper-evident event history, alerts and access reviews

3. Key user journeys

3.1 Merchant onboarding

• Create merchant workspace and administrator account.

• Select/import data sources and map required fields.

• Review data-quality and identity-match report.

• Configure messaging channel, consent source, quiet hours and frequency caps.

• Set approval thresholds and allowed actions.

• Preview initial customer segments and insights before enabling engagement.

3.2 Customer intelligence to action

• System refreshes customer/order/payment data.

• Intelligence layer identifies a segment or opportunity.

• Insight card shows reason, confidence, supporting records and freshness.

• Merchant reviews, edits, approves or rejects the proposed action.

• Policy engine checks consent, limits, channel rules and action permissions.

• Execution adapter sends or schedules the approved action.

• Delivery/response/order events are captured and attributed to the action.

3.3 Earned autonomy

• New action types start in recommendation or approval-required mode.

• System records merchant decisions and action outcomes.

• A merchant may configure a specific action type for limited automation after review.

• Only allowlisted actions inside merchant-defined thresholds may run autonomously.

• System pauses automation on policy violations, low confidence, unusual volume or integration errors.

• Merchant can revoke autonomy immediately; all decisions remain auditable.

4. Core domain model

Entity	Key attributes (illustrative)	Relationships / notes
Merchant	merchant_id, name, timezone, locale, settings	Tenant root; owns all business data
User / Role	user_id, merchant_id, role, permissions, status	Merchant-scoped access control
Customer	customer_id, merchant_id, identifiers, consent, preferences	May link to many orders and interactions
Order	order_id, customer_id, status, value, payment_mode, dates	COD/prepaid, fulfilment and return lifecycle
Payment / LedgerEntry	entry_id, customer_id, order_id, amount, due_date, status	Udhaar and collection history; authoritative source must be selected
Product / PurchasePattern	product_id, customer_id, cadence, last_purchase	Supports reorder estimation
Insight / Prediction	insight_id, type, score, confidence, reasons, model_version	Generated from versioned source data
Action / Approval	action_id, type, content, status, approver, policy_result	Approval and execution state machine
Message / Interaction	message_id, channel, template, delivery_status, response	Must retain consent and provider references
Experiment / Outcome	experiment_id, cohort, exposure, conversion, revenue	Supports incremental impact estimation
AuditEvent	event_id, actor, action, timestamp, before_after, correlation_id	Append-only audit history

5. Agent design requirements

Agent / service	Responsibility	Restrictions
Orchestrator	Select and coordinate specialist agents; manage workflow state	Cannot bypass policy engine or grant permissions
Customer Intelligence Agent	Summarise customer history and segments	Use only tenant-authorised data; cite source records
Behaviour / Prediction Agent	Estimate inactivity, reorder opportunity and COD risk	Return confidence, reason codes, model version and data freshness
Collections Agent	Prioritise due/overdue balances and draft reminders	No payment movement or balance mutation without approved integration/action
Engagement Agent	Draft localised, personalised messages	No send without consent and policy gate; respect opt-out
Policy / Safety Service	Evaluate permissions, consent, limits, quiet hours and risk rules	Fail closed when required evidence is missing
Execution Agent	Call approved external actions and record receipts	Allowlisted tools only; idempotent calls and bounded retries
Measurement Agent	Attribute exposures, responses and business outcomes	Do not claim causality without valid experiment design

6. UX and interaction requirements

• Dashboard: show sales/repeat purchase, customer reactivation, COD risk, collections and productivity indicators with date range and definitions.

• Customer 360: show identity confidence, purchase timeline, preferences, consent, outstanding balances and interaction history.

• Insight cards: display opportunity type, suggested next action, reason, confidence, evidence freshness and estimated impact where defensible.

• Approval inbox: provide approve, edit, reject, snooze and bulk review with clear action preview.

• Automation settings: configure per-action autonomy level, value/volume limits, quiet hours, daily caps and emergency pause.

• Audit and analytics: filter by actor, action, customer, date, channel, approval and outcome.

• Use plain language and Hinglish message previews where relevant; provide an English control-plane interface initially unless stakeholders decide otherwise.

7. Integration design

Integration area	Required interface behaviour	Open configuration
Commerce / orders	Import orders, customer identifiers, status and fulfilment events; reconcile updates	Initial platforms, API/webhook availability and sync frequency
Billing / ledger	Import invoices, due dates, payments and balance adjustments	System of record and supported ledger semantics
WhatsApp	Template management, consent validation, send API, delivery/read/reply webhooks and opt-out	Business provider, template approval and number ownership
Shipping / COD	Ingest shipment, delivery attempt, RTO and COD collection events	Carrier/aggregator and event taxonomy
Analytics	Export or query outcome events and experiment cohorts	Warehouse and BI choices

8. Security, privacy and governance

• Enforce tenant isolation at API, database query, background job, cache, search and object-storage layers.

• Use role-based access control and least-privilege service identities; require strong authentication for administrators.

• Encrypt data in transit and at rest; manage secrets in a dedicated secret store and rotate credentials.

• Maintain data provenance, purpose/consent records, retention schedules, deletion workflows and access logs.

• Minimise sensitive customer data sent to language models; redact or tokenise identifiers where practical.

• Do not expose one merchant's customer information, prompts, embeddings, analytics or model artefacts to another merchant.

• Require policy-engine approval before external side effects; log prompt/model versions and tool calls.

• Complete legal, privacy and security review for applicable Indian laws and provider terms before production.

9. Reliability and operations

• Use queues for asynchronous ingestion, prediction and messaging; support retries with exponential backoff and dead-letter handling.

• Make external writes idempotent and correlate every request, approval and provider receipt.

• Monitor connector freshness, job failures, queue depth, model drift, send failures, unusual action volume and policy denials.

• Provide manual fallback and an emergency pause for all autonomous execution.

• Back up critical data and test restoration; define recovery objectives with stakeholders.

10. Design decisions still open

• Cloud/provider, region, database, event bus, vector/search technology and analytics stack.

• Initial integrations and source-of-truth ownership for customer, order and ledger records.

• Identity resolution rules and acceptable match confidence.

• Model approach, evaluation dataset, risk thresholds and retraining cadence.

• Agent framework and model provider; approved tools and data-sharing terms.

• WhatsApp provider, template ownership, language variants and opt-out workflow.

• Exact retention, deletion, backup, availability and disaster-recovery targets.

SRS — Software Requirements Specification

SELLVION

Software Requirements Specification

SRS | 0.1 — Draft for review | 12 November 2026

Source basis: three user-provided SELLVION concept screenshots.

Document status	Draft — requires stakeholder validation
Product	SELLVION — AI customer intelligence and engagement platform
Audience	Business stakeholders, product/design, engineering, QA and AI implementation teams

1. Introduction

This SRS defines proposed software behaviour for the SELLVION product concept. Requirements are testable where possible. Any numeric service levels, thresholds or external integration details not provided by the screenshots are marked TBD and must be agreed before implementation.

2. Product perspective and user classes

• Merchant Admin: owns workspace, integrations, users, consent configuration and automation policy.

• Merchant Operator: views customers and insights, reviews approvals and handles exceptions.

• SELLVION Support/Admin: operates platform support under audited, restricted access.

• End Customer: interacts through approved messaging channels; does not need a SELLVION account for basic message interactions.

3. Functional requirements

ID	Feature	Requirement	Priority
FR-001	Tenant setup	The system shall create a separate merchant workspace with a unique merchant identifier.	Must
FR-002	User access	The system shall authenticate users and enforce merchant-scoped role permissions.	Must
FR-003	Data source setup	The system shall allow an authorised admin to connect, import or map supported customer/order data sources.	Must
FR-004	Data provenance	The system shall record source, ingestion time, source record ID and sync status for imported records.	Must
FR-005	Customer identity	The system shall link customer records using configured identifiers and retain a match-confidence/status indicator.	Must
FR-006	Customer profile	The system shall present a merchant-scoped customer profile containing available purchase, order, preference, consent and interaction history.	Must
FR-007	Segmentation	The system shall classify customers into configurable or system-defined segments such as inactive and regular buyers.	Must
FR-008	Reorder opportunity	The system shall generate reorder opportunities from purchase history and expose the reason and confidence.	Must
FR-009	COD risk	The system shall calculate or obtain a COD risk indicator when sufficient permitted data exists and expose its reason, confidence and model version.	Should
FR-010	Udhaar tracking	The system shall display outstanding and overdue balances from the configured source of truth, including due dates and source references.	Must
FR-011	Message drafting	The system shall generate personalised message drafts using permitted customer and business context.	Must
FR-012	Consent gate	Before sending a message, the system shall verify applicable consent, opt-out status, channel eligibility and provider requirements.	Must
FR-013	Approval workflow	The system shall support approve, edit, reject and cancel operations for actions requiring merchant review.	Must
FR-014	Policy limits	The system shall enforce merchant-configured action allowlists, value/volume limits, quiet hours and frequency caps.	Must
FR-015	Message execution	The system shall submit an approved message through a configured provider and record provider response and delivery status.	Must
FR-016	Autonomy levels	The system shall support recommendation-only, approval-required and bounded-autonomous modes per action type.	Must
FR-017	Autonomy stop	The system shall pause an autonomous action type when a configured safety threshold, policy violation or critical integration error occurs.	Must
FR-018	Audit trail	The system shall record actor, timestamp, action, approval, policy result, model/prompt version and execution result for consequential actions.	Must
FR-019	Outcome tracking	The system shall capture message engagement, customer response and subsequent order/payment outcomes where data is available.	Must
FR-020	Revenue measurement	The system shall support cohort/holdout reporting and label estimated incremental revenue separately from observed revenue.	Must
FR-021	Dashboard	The system shall display KPI summaries with date range, metric definitions and data freshness.	Should
FR-022	Opt-out	The system shall record and enforce customer opt-out requests across relevant messaging workflows.	Must
FR-023	Exception handling	The system shall surface failed syncs, unmatched identities, missing consent, provider errors and uncertain predictions for review.	Must
FR-024	Export	The system should allow authorised users to export permitted reports and records with audit logging.	Could

4. External interface requirements

4.1 User interface

• Responsive merchant console with dashboard, customer list/profile, insights, approval inbox, automation settings, integrations, audit and reports.

• All actions with external side effects must show a preview and a clear status before execution.

• Display missing, stale or low-confidence data explicitly; never present unavailable data as zero.

• Provide accessible labels, keyboard navigation, readable contrast and clear error messages.

4.2 Software interfaces

• Commerce/order connector interfaces shall support authenticated import and/or webhook ingestion with documented field mapping.

• Messaging adapter shall support approved templates, send requests, provider message IDs and delivery/reply/opt-out callbacks.

• Ledger connector shall provide balance, due date, payment and adjustment events from the selected system of record.

• All connectors shall expose health, last successful sync, error state and retry status.

4.3 Communication interfaces

External APIs shall use authenticated encrypted transport. Webhooks shall be authenticated, replay-protected where supported and processed idempotently. Payload schemas and provider-specific limits are TBD until providers are selected.

5. Non-functional requirements

ID	Requirement	Target / status
NFR-001	Tenant isolation shall prevent cross-merchant access to data, files, logs, caches and model context.	Mandatory; verify by security tests
NFR-002	Data shall be encrypted in transit and at rest.	Mandatory; algorithm/key policy TBD
NFR-003	System shall maintain an auditable record of privileged access and consequential actions.	Mandatory
NFR-004	The system shall support horizontal scaling of ingestion and asynchronous jobs.	Design requirement
NFR-005	The system shall recover safely from transient connector/provider failures without duplicate external actions.	Idempotency and retry tests required
NFR-006	User-facing pages shall provide agreed response-time targets under agreed load.	TBD by product/engineering
NFR-007	Availability, RPO and RTO shall be agreed before production launch.	TBD
NFR-008	The system shall support retention, export and deletion policies approved by legal/privacy stakeholders.	Mandatory; durations TBD
NFR-009	AI outputs shall include confidence/uncertainty and traceable reason codes where applicable.	Mandatory for predictive recommendations
NFR-010	Critical automation shall fail closed when consent, permissions or policy evidence is unavailable.	Mandatory
NFR-011	The interface shall be usable on current supported desktop and mobile browsers.	Browser matrix TBD
NFR-012	Logs and telemetry shall avoid unnecessary exposure of personal/customer content.	Mandatory

6. Business rules

• BRULE-01: A customer record and all related data must be scoped to exactly one merchant tenant unless an explicitly approved shared-data design exists.

• BRULE-02: No customer message may be sent if consent is missing where required, the customer has opted out, the channel is ineligible or the provider rejects the template.

• BRULE-03: Recommendation confidence below the configured threshold must require human review or be suppressed.

• BRULE-04: Autonomous execution is permitted only for action types explicitly enabled by the merchant and within configured limits.

• BRULE-05: A missing or stale ledger source must be shown as unknown/stale, not as a confirmed zero balance.

• BRULE-06: Incremental revenue is an estimate derived from a valid comparison method and must be distinguished from directly observed sales.

• BRULE-07: COD risk scores must not be represented as certainty or used for prohibited discriminatory decisions.

7. Core workflows and states

Object	Illustrative states / transitions
Insight	Generated → Available → Accepted / Dismissed / Expired
Action	Draft → Policy Check → Approval Required → Approved → Queued → Executed → Outcome Recorded; alternate states: Rejected, Cancelled, Failed, Blocked
Message	Draft → Validated → Submitted → Sent / Failed → Delivered / Undelivered → Read / Replied / Opted Out (provider-dependent)
Connector sync	Pending → Running → Succeeded / Partial / Failed → Retried
Autonomy policy	Disabled → Recommendation Only → Approval Required → Bounded Autonomous → Paused

8. Acceptance criteria (initial)

• An admin can create a merchant workspace and invite a second user with restricted permissions; the second user cannot access another tenant.

• An imported customer/order record retains source identifiers and ingestion timestamp and appears in the correct customer profile.

• A duplicate or uncertain identity match is flagged for review rather than silently merged.

• A reorder insight displays the evidence window, confidence and recommended next action.

• A message cannot be sent when consent is absent/invalid or an opt-out is recorded.

• An approval-required action cannot execute until an authorised user approves it.

• An autonomous action outside configured limits is blocked and logged.

• A transient provider retry does not create duplicate messages.

• Dashboard reports show metric definitions, selected date range and data freshness.

• Revenue reporting labels estimates and shows cohort/control methodology.

9. Data requirements and validation

• Minimum customer fields: merchant_id, source_customer_id or approved matching identifiers, created/updated timestamps and consent status/source where applicable.

• Minimum order fields: source_order_id, customer reference, order date, value/currency, payment mode, order status and fulfilment/RTO status when available.

• Minimum ledger fields: source_entry_id, customer reference, amount, due date, status, currency and last-updated timestamp.

• Every field mapping shall define type, required/optional status, validation, null handling and source precedence.

• Data retention, deletion, correction and export requirements must be confirmed before production.

10. Test requirements

• Unit tests for validation, segmentation, policy rules, state transitions and metric calculations.

• Integration tests for connector authentication, pagination, webhooks, retries, idempotency and provider error handling.

• Security tests for tenant isolation, role permissions, secret handling, injection and audit integrity.

• AI evaluation for precision/recall by segment, calibration, false-positive/false-negative cost, language quality and drift.

• End-to-end tests for onboarding, insight review, approval, message delivery, opt-out, exception handling and outcome reporting.

• Load and resilience tests at agreed merchant, customer, event and message volumes.

11. Traceability to business objectives

Business objective	Related functional requirements
Revenue growth / repeat purchases	FR-006, FR-007, FR-008, FR-011, FR-019, FR-020, FR-021
Customer retention	FR-006, FR-007, FR-011, FR-019
COD risk reduction	FR-009, FR-023
Faster collections	FR-010, FR-011, FR-015, FR-019
Higher productivity	FR-013, FR-014, FR-016, FR-017, FR-021, FR-023
Personalised experience	FR-006, FR-011, FR-012, FR-022
Smarter decisions	FR-008, FR-009, FR-020, FR-021

12. Unresolved items

• Confirm exact meaning of DRD and whether a separate data requirements document is also needed.

• Confirm first-release scope and all named external systems/providers.

• Set measurable latency, availability, scale, recovery, model-quality and operational targets.

• Approve consent, retention, deletion, privacy and legal policies.

• Define exact COD/reorder model inputs, thresholds and human escalation rules.

• Define financial attribution methodology, experiment design and reporting window.

• Define message templates, languages, quiet hours, rate limits and supported customer channels.